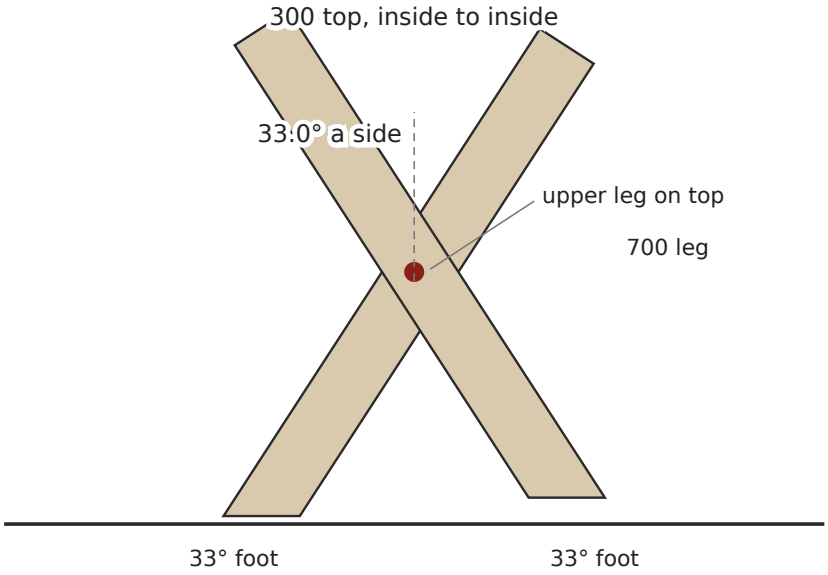
X-Leg Herb Trough

your build · v3

- Your numbers. Two X-leg frames carry a 900 long trough between them.
- Legs **4×700**, one end cut 33°. Cross planks **8×900**, four front and four back. Planks 80 wide × 23 thick.
- **The X is set by the top.** Both leg tops sit **300 apart, inside corner to inside corner** - that is your gauge, and you cut to it so every X matches.
- The two legs do not notch: the upper leg lies straight **on top of** the lower one, so there is an inside leg and an outside leg.
- The sides are not cut to a number. They are **scribed off an offcut** held in the V, one on top of the last.
- Shown at 900 long, four planks deep. Dimensions in mm.

X JOINT - stage 1



Stage 1 - the joint that decides the box. Two 700 legs crossed at their centres, upper leg lying on the lower, top set to 300. Get the foot flush on the clamped plank and the top to 300 and the frame stands true - and both frames match.

CUT LIST

PART	QTY	LENGTH	WIDTH	THICK	NOTE
Leg	4	700	80	23	one end cut 33°, all four one setting
Cross plank (wall)	8	900	80	23	4 front, 4 back
Side piece	6	scribed	80	23	3 per end, marked off an offcut
- of those, the bottom pair	2	900	80	23	first plank's inner long edge cut 22.5° into the V; second trimmed 23 mm

MATERIAL

- Untreated pine or hardwood. Untreated: it is a food bed.
- **35 mm part-threaded screws** for the crossing and the side courses.
- A **6 mm or 10 mm bolt** through each crossing if you want it to never move.
- Stainless or coated only. Wood oil (Woodoc) outside and on the feet.

Two rules that decide this build

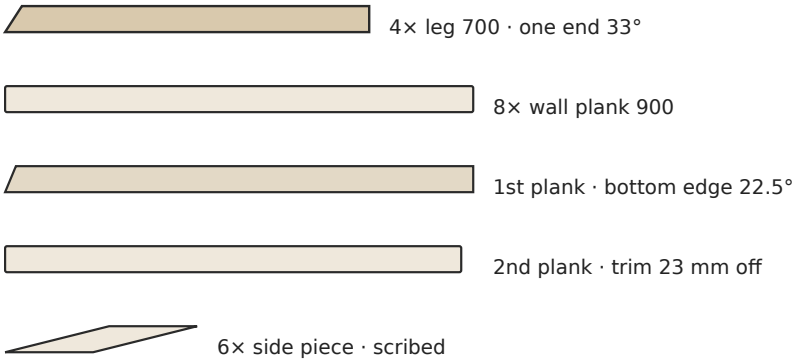
- **One saw setting for all four feet.** Cut every 33° end off the same stop. One foot cut on its own and the box stands on three legs.
- **Lock the crossing.** A single screw lets the X scissor. Add 3-4 screws, or bolt it.

Locked in from your build photos (14 Oct)

- **Top = 300, inside corner to inside corner.** Your gauge to check every X comes out the same.
- **Feet cut at 33°** so they sit flat. Method: clamp a plank, leg flush, one centre screw into the middle of the plank.
- **The two legs cross one on top of the other.** No notch - the upper leg lies full-thickness on the lower, so there is an inside leg and an outside leg.
- **Spacer blocks at the shoulder.** Small blocks fill the gap between the angled leg head and the underside of the first planks. Drawn from the ~60 and ~75 mm marks in your shots.
- **One wall flush, one proud.** The back planks run **flush to the outside leg**; the front planks stand **23 mm proud** (one plank thickness), as in your shots.

Cut list, drawn what to cut

Every cut in the build. Two angles hold the whole thing: **33°** on the leg, **22.5°** in the V, and the rest square.

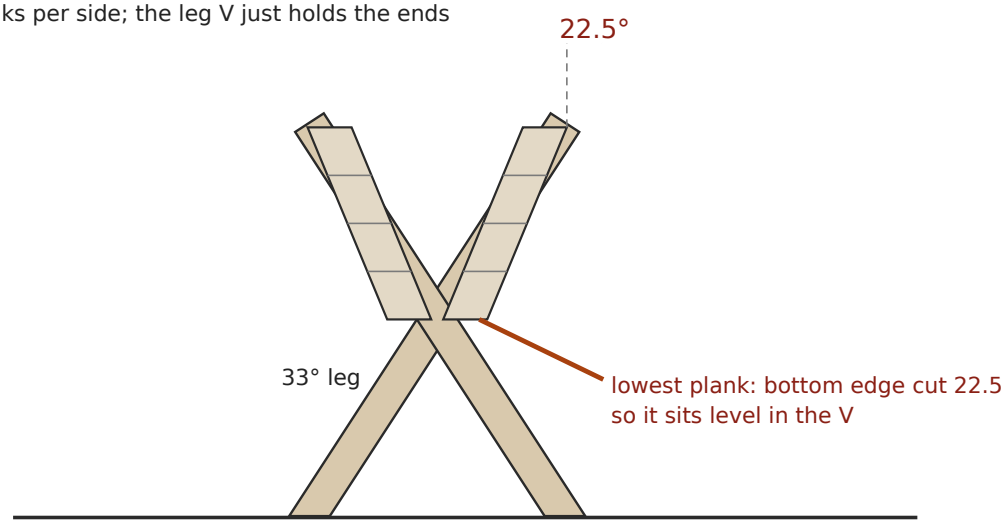


THE 22.5° V

THE 22.5° V - stage 2

the box is its own V - two sides leaning in, meeting at the bottom

4 planks per side; the leg V just holds the ends



The box is built into the leg V as its own V. Both long sides lean in at 22.5° and close at the bottom. The lowest plank's bottom edge is cut at 22.5° so it sits level; the second lands one thickness (23 mm) higher, so trim 23 mm off its bottom edge and the tops come level.

The two angles that fool everyone

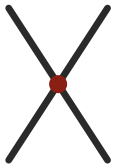
- 33° is the **leg** - the foot cut and the lean, one saw setting. Nothing else is 33°.
- 22.5° is the **V** - how far each side leans in, and the cut on the lowest plank's bottom edge to match it.
- Each side above that is **scribed** off an offcut stood in the V, not set to a number.

The build in four stages

picture sequence

Four stages, in order. Do not start the sides until the walls are screwed down and level.

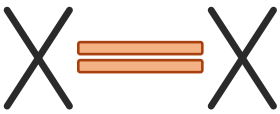
1



33° feet · screw the centre

Stage 1 - the X legs. Cut one end of each 700 leg at 33°. Cross two: the upper leg lies straight **on top of** the lower one. Centre 350 along, 40 across, one screw through the middle. Make two, mirrored - one set inside, one outside, both at the same 300 top.

2



first two in the V

Stage 2 - stand them and start the floor. Stand the two X sets parallel, 300 top. Cut one **long edge** of the first 900 at **22.5°** - the inner edge that lands on the far leg - so it beds into the V. The second lands one thickness (23 mm) higher, so trim 23 mm off its edge so both tops come level.

3



4 front + 4 back

Stage 3 - the rest of the 900s. 4 planks on the front, 4 on the back, flush with the left and right X. The back sits 23 mm proud behind the outside leg. Screw down front and back.

4



sides off an offcut

Stage 4 - the sides. Hold an offcut in the V, mark the angle, cut the first side piece. Next piece higher: mark off the first, cut, glue and screw on top. Third the same. Mirror the other end.

Stage by stage 1 – 2

#	STAGE	WHY
1	**The X legs.** • Four legs, 700 long. Cut **one end of each** at 33° - all four off the **same saw setting**. • Find the centre of each leg: 350 along, 40 across (half of 80). • Clamp a straight plank to the table. Press the 33° end flush against it. • Lay the second leg **straight on top of the first** to make the X. Pilot hole at the centre. • Drive a 35 mm part-threaded screw down through both at the centre so the upper leg sits on the lower one. • Then 3-4 more screws, or swap to a 6 mm or 10 mm bolt through the crossing. • Set the **top to 300, inside corner to inside corner** - that is your gauge. Make the second set the same.	• One saw setting for all four feet = feet that sit level. Cut them one at a time and they fight. • The clamped plank is the reference - it stops the X leaning while you screw. • The legs do not notch into each other: one lies full-thickness on top of the other. That gives an inside leg and an outside leg. • A single screw lets the X scissor. Extra screws or a bolt lock the crossing. • 300 top on both sets = the box sits level and the walls line up.
2	**Stand them and lay the first two planks.** • Stand the two X sets parallel, on the ground. • Lay the first two 900 planks in the V, one front, one back. • Cut **one long edge of the first plank at 22.5°** - the inner edge where it meets the far leg - so it beds into the V. • The second plank lands one thickness (23 mm) higher, so **trim 23 mm off** its edge so both tops come level.	• Both X sets parallel = the box goes up straight. • The 22.5° on the inner edge is what lets the plank bed into the V instead of rocking on the leg corner. • Level first planks = level box. This is the one that sets it.

Stage by stage 3 – 4

#	STAGE	WHY
3	**Lay the rest of the 900s.** • Carry on stacking: 4 planks on the front, 4 on the back, one on top of the other. • All flush with the left and right X planks. • The back stacks 23 mm proud, behind the outside leg. • Both sides finish at the same height. • Screw the stack down, front and back, into the X frames.	• Flush to the X = straight walls, no stepping. • The back offset clears the leg, so both walls meet the frame the same way. • Equal height both sides = a level rim.
4	**The sides, scribed off an offcut.** • Hold an offcut horizontal in the bottom of the V. Draw a line from inside to mark the angle of the first side piece. • Cut to the line, fit it in the V. • Hold a longer offcut above the first, mark off it, cut the second, glue and screw it on top of the first. • Third the same, on top of the second. • Do the same for the opposite end.	• The V is not a set angle you can measure off a drawing - you scribe each piece off the one below. • Glue **and** screw: end grain is weak, the glue takes the load, the screws hold it while it sets. • Three pieces fill the end from the floor to the rim.

Your build, 14 Oct

reference

The four shots you sent. These are the ones the drawings answer to.



The X joint. The upper leg lies on top of the lower - no notch. One screw through the centre.



The foot. The 33° cut sitting flat, and the crossing above it with the centre screw in.



The floor. First planks laid across, sitting in the V of the two frames.



Top view of the set. Crossed legs on the tray, tape marks where you set the spread and the angle.